Predictive Analysis of Crime Rates Using Machine Learning Algorithms

Jitendra Kumar Gupta[1,2]
1. Uttaranchal Institute of Technology, Uttaranchal University, Dehradun-248007,Uttarakhand, India
2. Department of Computer Science & Engineering ,Dr. BR Ambedkar National Institute of Technology Jalandhar-144011, India
jk760429@gmail.com

Mukesh Kumar Jha
Galgotias University
Greater Noida, India
mukeshjha84@gmail.com

Neeraj Rathore
Operations and Technology Management Cluster
School of Business, UPES
Dehradun, UK, India
navneer.rathorez@gmail.com

Priyanka Suyal
COER University
Roorkee, India
Priyankasuyal48@gmail.com

Srivaramangai R, Department of Information Technology, University of Mumbai, Mumbai, India
rsrimangai@gmail.com

Sakshi Tewari, Assistant Professor, Symbiosis Law School, Noida, Symbiosis International (Deemed University), Pune, India
Sakshiji.tewari@gmail.com

Kapil Joshi
Department of CSE, Uttaranchal Institute of Technology, Uttaranchal University, Dehradun-248007,Uttarakhand, India
Kapilengg0509@gmail.com

2024 4th International Conference on Innovative Sustainable Computational Technologies (CISCT) | 979-8-3503-7814-6/24/$31.00 ©2024 IEEE | DOI: 10.1109/CISCT62494.2024.11134190

***Abstract*— This study outlines a leader-edge technique that machine learning introduces in crime prediction over traditional models. In the face of intensifying urbanization and escalating population growth which is completely changing the societies, it is a core of grasp and have a foresight of crime patterns that law enforcement agencies and the policymakers should focus on. This study captures diverse features comprising demographic data, historical crime records, socioeconomic indicators, and geographic information among others, to develop comprehensive data inputs into the advancements in machine learning algorithms which in turn compute the predictive models. The technique comprising of data preprocessing, feature engineering, and model building is using a plethora of algorithms including decision trees, random forests, support vector machines, and neural networks. Another aspect of paper is to explain why it is important for feature selection and hyperparameter tuning in improving the model's performance.**

***Keywords*— Machine Learning, KNN, SVM, Crime Rate, Random- Forest.**

I. INTRODUCTION

The crime is doing what is prohibited by the law and may result in punishment, such as fines, imprisonment or more legal action. It is an element of society which dates back to primordial times and remains one of the major concerns even now with the advent of technology and extension of globalization. Crime pattern predictions is the very necessity for considering whether crime rates are going to increase or decrease. Machine learning, discipline which is dealing with algorithms and data driven methods participates in the detection and forecasting of criminal activity. Based on analysis of the crime patterns including demographic data and environmental factors machine learning can generate models useful to estimate the probability of criminal phenomena occurrence in different locations and time periods [1]. Additionally, machine learning yields insights into behavior allowing officers to respond and come up with measures of countering activities rationally. Crime is a security challenge because growth of the populations in urban areas entails criminal activities. Authorities start to consider prediction of crime rates based on a huge amount of data. With the use of models based on information of historical crimes actual places where any criminal act might happen can be found prior in advance and so the efforts of preventing crime and improving strategies of law enforcement can be bettered [2]. The purpose of this initiative is to design and implement a viable mechanism that can accurately predict crime patterns and spot trends. This will give the authorities the mandate to come up with intervention initiatives that will be aimed at reducing the crime rate and creating an environment explains how society settles on the choice of a detention rule. present a novel approach for rent forecasting, which is focused on the US housing market. The contribution is three-fold: machine learning approach is used with online data resources such as Airbnb, Zillow, Schools, Public transportation, and crime rate websites: The system will apply the machine learning algorithms to forecast crime rates using the factors of year, location, and type of crime as its inputs [3].

Table 1 showing crime rate in various country [3] Furthermore, machine learning can be used for spotting and discovering the patterns and correlations among varieties of crimes, the previous which will improve the distribution of resources to the maximum. With the ability of computers to process large volumes of data and flag any departures or potential future crimes, officers gain extra

979-8-3503-7814-6/24/$31.00 ©2024 IEEE

Authorized licensed use limited to: Pontificia Universidad Catolica Del Peru. Downloaded on September 18,2026 at 14:37:41 UTC from IEEE Xplore. Restrictions apply.

Country	Crime rate (%)
Venezuela	83.76
Papua New Guinea	80.79
South Africa	76.86
Afghanistan	76.31
Honduras	74.54
Guyana	68.74
El Salvador	67.79
Brazil	67.49
Jamaica	67.42
Syria	67.18

TABLE I. Crime Rate in various country

time needed for being in the right place at the right time and preventing criminal activities to take place.In addition, machine learning can be leveraged to analyze social media and other online forums in order to collect information about emerging criminal activities (criminal activities as they are happening). Agencies of security at the help of tracking online networking second and the related posts, may spot the apporochement of possible threats and take proper action to prevent the happening of the crime.In most cases where the crime predictions and the crime prevention processes of law enforcement agencies are considered, merging machine learning can be pushed forward as a possible futuristic measure. Without going through the stages that other criminal organizations have to deal with, the current crime syndicates have utilized data structure as well as algorithms to stay one step ahead of law enforcement have at their disposal a complete safe and protected community for everybody [4].

II. LITERATURE REVIEW

The "reality" of the prediction tasks coupled with the amount of data available implies that this is a good machine-learning application. Besides, comparing the algorithm to judges is rather complicated [4]. This paper is looking at the crime prediction in YD county by the use of data mining and machine learning. The aim of is displaying the crime pattern and rate in the YD county by using the data gathered and to show the relationship that exists among the various character and character Variable [5]. investigate the violence that occurred in the urban community, sifting through police crime reports and demographic data for Chicago during 16 years. Usefulness of more than тс months provides a highly accurate prediction result, on the other hand, it is harmful to decrease [6] the amount of months to be used. It's a matter of lively question whether Machine Learning algorithms used in the meaningful areas (for example, criminal justice) treat people of different social groups "fairly." Let's take criminal justice as an example (Jung et. Moreover, (Hasbrouck et various machine learning algorithms, logistic regression, SVM, Naïve Bayes, KNN, decision tree, MLP, random forest, and XGBoost along with time series models such as LSTM and ARIMA in order to be perfectly complement the crime data [7]. Many prediction [9] methodologies have been suggested to aid the machine learning crime prediction problem India be in Introduce the useful approach assemble-stacking crime pre- diction method(SBCPM) to predict crimes exactly by using learning-based methods, and the implementation of MATLAB In order to overcome cyber-crime and criminals, there are two very important components: the first one being identification of the attackers and the second one being how they attack.

forecast the impact of the specified features on the identification of the cyber criminals method and perpetrator. Machine Learning algorithm on crime data may lead to crime prediction at an area level [8]. approach to police work concerning crime detection incorporates machine learning based soft computing regression analysis methods. Accordingly, an approach of Machine Learning-based soft computing regression approach for Indian Crime Data Analysis (ICDA) is proposed. Yet, there are many aut Crime rate prediction using machine learning has been a focal point in recent research efforts. Several studies have investigated the utilization of machine learning techniques to improve the accuracy and dependability of crime prediction systems [9]. Researchers have developed pre- diction models employing various machine learning algorithms such as Linear Regression, SVM, KNN, decision trees, Naive Bayes, and CART to effectively forecast crime rates. Machine learning models have proven successful in predicting criminal justice outcomes, particularly in the context of place-based crime predictions [10].

Furthermore, the integration of machine learning and visual-ization tools has played a crucial role in predicting crime rates and identifying high-risk crime areas. Some stud- ies have concentrated on creating ensemble learning models capable of predicting spatial crime occurrences at a detailed level without relying on historical crime data . Additionally, machine learning techniques have been used for crime data analysis with the aim of reducing crime rates and deterring criminal activities [11]. Moreover, supervised and unsupervised machine learning methodologies have been applied for crime pattern analysis, demonstrating the versatility of these techniques in under- standing and predicting criminal behaviors . The amalgamation of big data approaches with ma- chine learning has displayed potential in enhancing crime detection and prediction accuracy [12]. Additionally, the exploration of stacked generalization and ensemble

approaches has been undertaken to enhance the efficacy of machine learning algorithms in crime prediction. In the domain of crime prediction, machine learning models have been employed for various purposes, including recidivism prediction, hate crime analysis, cyber-attack prediction, and child abuse prediction [13]. These diverse applications underscore the adaptability of machine learning in addressing different aspects of crime prediction. Overall, the literature review emphasizes the increasing interest and effectiveness

Authorized licensed use limited to: Pontificia Universidad Catolica Del Peru. Downloaded on September 18,2026 at 14:37:41 UTC from IEEE Xplore. Restrictions apply.

of machine learning in predicting crime rates and improving law enforcement strategies.

Here are the top ten countries with highest crime rate per 100k in 2023 [14] mention in Table 2:

TABLE II. Murder crime rate per 100,000

Country	Murder rate
Celaya, Mexico	109.39
Tijuana, Mexico	105.15
Ciudad Juarez, Mexico	103.61
Ciudad Obregon, Mexico	101.13
Irapuato, Mexico	94.99
Ensenada, Mexico	90.58
St.Louis, United States	87.83
Uruapan, Mexico	72.59
Fiera De Santana, Brazil	67.46
Cape Town, South Africa	64
Cumana, Venezuela	62.42

Reducing crime rate via machine learning (ML) has been the area of concern of recent researchers. Many studies have been resulted in the use of machine learning algorithms for the purpose of increasing the accuracy and reliability of the crime prediction systems [15]. Machine learn- ing algorithms like Linear Regression, SVM, KNN, decision trees, Naïve Bayes and CART are among the prediction models being developed by researchers to better classify crime . Machine learning techniques have been found to be good in the sense that they have been able to correctly predict criminal justice outcomes especially in the situation of place-based crime prediction [17]. In addition, the capabilities of machine learning and visualizing devices in forecasting crime rates and determining crime hots feel was also very important. Others have been concerned with pertinent sub models which would predict spatial crime occurrences using fine- grid patterns without having to rely on historical crimes data. As for the application of machine learn- ing techniques, they are also known for crime data analysis aiming at the reduction of crime rates and crime deterrence [18]. Additionally, the supervised and unsuper- vised machine learning techniques were employed for crime pattern analysis. This shows the adaptability of these methods to apprehend criminal behaviors. The use case of big data methodology integrated with machine learning has unraveled a great deal of promise in the area of crime detection and prediction accuracy [18]. Furthermore, the exploration of stacked generalization and ensemble approaches has been carried out and looked at this year in order to increase the success rate of machine learning algorithms in crime prediction [19,20].

III. METHODOLOGY

Ensemble learning is a method in machine learning combining many models together to maximize predictive accuracy [21]. It does so by developing systems from the same training set that are composed of various datasets and chips [22]. Those models can generate more accurate projections. This method places different data sets in a single joint space, leading to its use in multiple areas such as computer vision and natural language processing [23]. Random Forest is a combination of multi- deciders, which can be trained using the error-correcting mechanisms of decision trees, to generate better predictions. It is normally used for variance reduction and also sustaining prediction accuracy through the assumption of randomly sampling data, only selected features, and several decision trees on data. The trees then combine their predictions, less the biases of each algorithm by utilizing voting to attempt to arrive at the final prediction. Random Forests are more precise than, for instance, original Decision Trees and remain stable even in cases of complex tasks dealing with multitudes of data. The method consists in building an algorithm that starts with a dataset of observations and labels at random, letting for 'k' features, finding the best split point, making a tree with the chosen split points, repeating steps 2, 3, 4 for each trait, combining the trees, that is having new data predicted by them and then accuracy of the predictions determined.

In the format that is suitable for analysis, several columns might disappear and transform, or labeling is the method employed to convert categorical data into numeric for better prediction. Random Sampling: The data set has been split into two parts after the feature selection process: the training data that covers 70% of the data and the testing data set that is around 30% of the data. Model Creation: The model algorithms in this project are borrowed from sklearn. Hence, model is created from model. Fit(). The dataset has been analyzed using five different models: support vector machine, nearest neighbor, decision tree, random forest, and artificial neural networks. Model Selection: Model performance along with the stated goals determined random forest model to be the one we have chosen. Class prediction is done using model. Predict(). Scoring accuracy is controlled by using the accuracy score which is imported from the metrics. Model Deployment: The model has been deployed for prediction using various web technologies (the cloud computation platforms, API, and AJAX).

IV. PROPOSED SYSTEM

At the inception, the data set is generated manually based on publication will be accessed from the National Crime Rate Bureau (NCRB) website. The method entails gathering data from a source and separating it into training and testing sets. Various machine learning techniques, such as Linear Regression, Support Vector Machines, Logistic Regression, and KNN Classifier, are taught and evaluated. The best-performing model is chosen for scoring and used to forecast crime rates using application data shown in Figure 1,2 and 3.

Authorized licensed use limited to: Pontificia Universidad Catolica Del Peru. Downloaded on September 18,2026 at 14:37:41 UTC from IEEE Xplore. Restrictions apply.

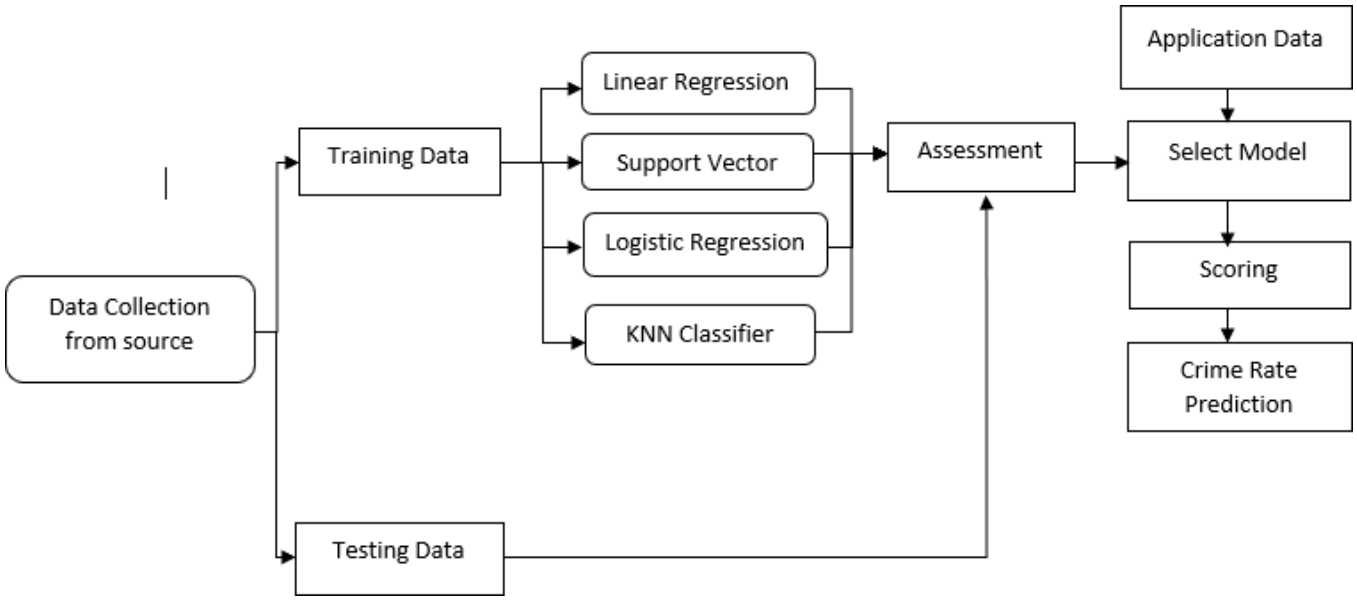


Fig.1. Proposed Machine Learning Workflow for Crime Rate Prediction

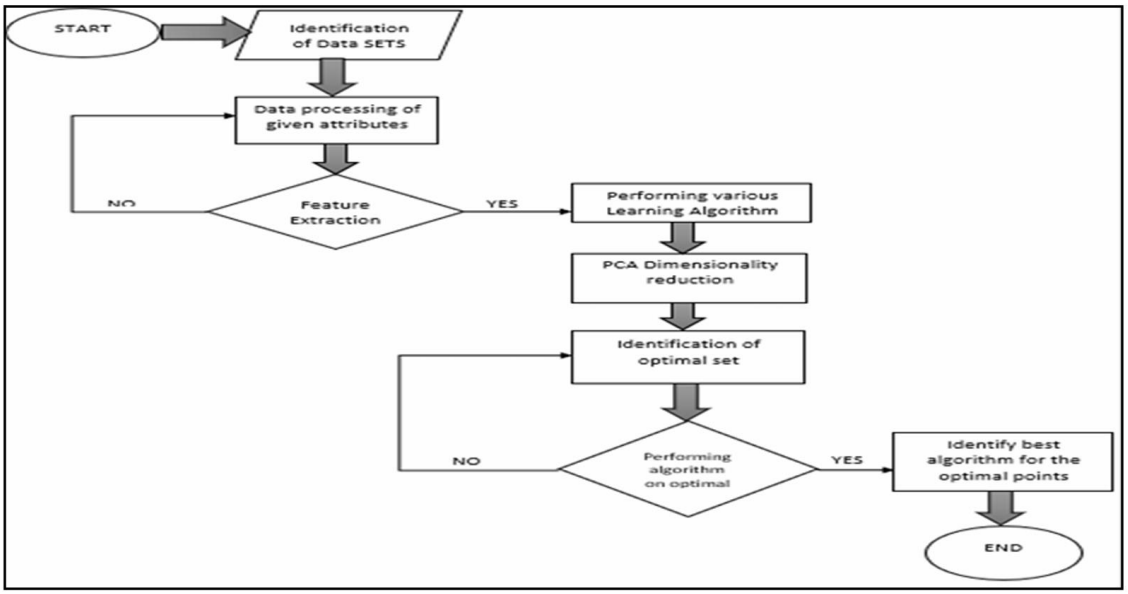


Fig.2. Feature Extraction and Algorithm Selection Process for Crime Rate Prediction

Authorized licensed use limited to: Pontificia Universidad Catolica Del Peru. Downloaded on September 18,2026 at 14:37:41 UTC from IEEE Xplore. Restrictions apply.

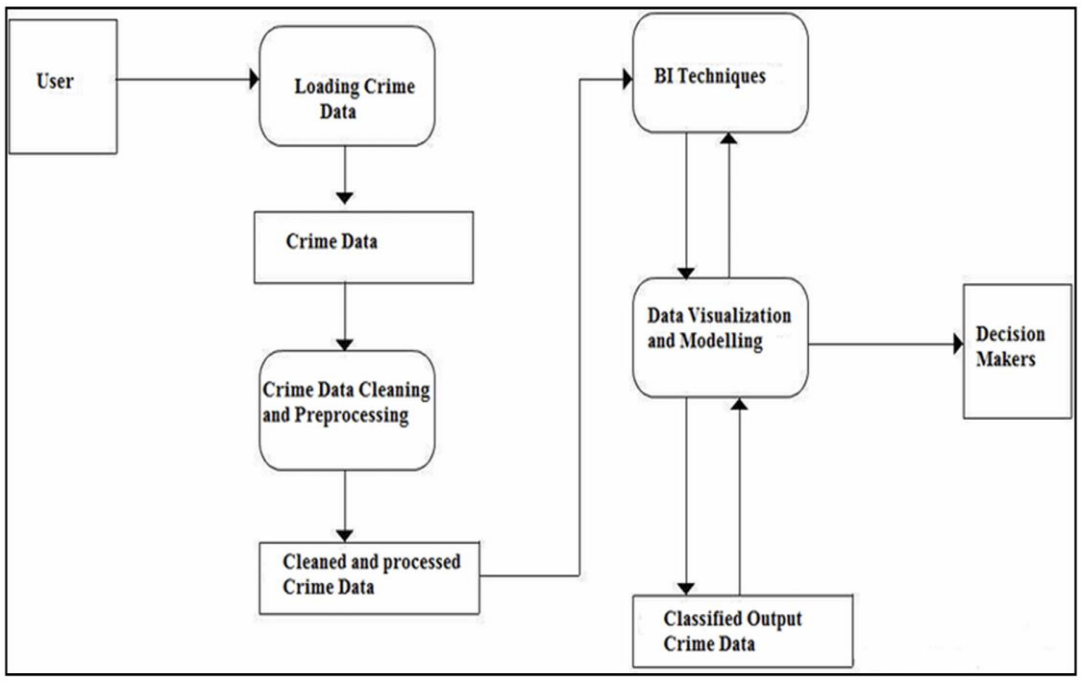


Fig 3 shows block diagram and system architecture of our proposed model and its working.

V. RESULT AND DISCUSSION

The Random Forest Regression model demonstrates the best accuracy in predicting test data among the five selected models. The model predicts the crime rate value for 10 different categories of crimes, including Murder, Kidnapping, Crime against Women, Crime against Children, Crime Committed by Juveniles, Crime against Senior Citizens, Crime against SC, Crime against ST, Economic Offenses, Cyber Crimes that will occur in 19 Indian metropolitan cities: Ahmedabad, Bengaluru, Chennai, Coimbatore, Delhi, Ghaziabad, Hyderabad, Indore, Jaipur, Kanpur, Kochi , Kolkata, Kozhikode, Lucknow, Mumbai, Nagpur, Patna, Pune, Surat in future. The accuracy results obtained after testing are listed below in Table 3:

Table III Outcome of the proposed model

Algorithm	**Mean Absolute Error**	**Mean Squared Error**	**R2 Score**
Support Vector Machine	10.3204	371.7907	0.17886
K-Nearest Neighbor	6.58181	140.8179	0.55349
Decision Tree Regressor	2.89024	34.95932	0.88915
Random Forest Regressor	2.49143	21.43956	0.93201

VI. CONCLUSION

Predicting crime rates has grown in importance as a strategy for law enforcement organizations looking to concentrate their efforts on high-crime areas. The timing and location of crimes can be predicted by law enforcement organizations with the use of advanced algorithms and data analysis. Through targeted resource allocation, law enforcement officials can contribute to a decrease in the general crime rate within a community. It appears that predictive policing will remain a crucial tool in the future, as it has already shown to be a beneficial tool in many places for lowering crime rates. Machine learning technology has made it simpler to identify connections and patterns within diverse data sets.

project's main goal is to forecast the future crime rate based on the year, the city, and the kinds of crimes that will occur. Using the machine learning principle, the training data has been cleaned and modified to generate a machine learning model. The accuracy of the model's crime rate prediction is 93.20%. The examination of the data set and the prediction of crimes are aided by the data visualization and crime rate prediction model. Numerous graphs are made in an effort to uncover intriguing statistics that aid in the comprehension of various crime datasets that are useful in putting into practice the elements that can contribute to maintaining societal safety.

Authorized licensed use limited to: Pontificia Universidad Catolica Del Peru. Downloaded on September 18,2026 at 14:37:41 UTC from IEEE Xplore. Restrictions apply.

References

[1] Butt, U., Letchmunan, S., Hassan, F., Ali, M., Baqir, A., & Sherazi, H. (2020). Spatio-temporal crime hotspot detection and prediction: a systematic literature review. Ieee Access, 8, 166553-166574. https://doi.org/10.1109/access.2020.3022808

[2] B˙ILEN, A. and Özer, A. (2021). Cyber-attack method and perpetrator prediction using machine learning algorithms. Peerj Computer Science, 7, e475. https://doi.org/10.7717/peer j-cs.475

[3] Espinosa, F. (2020). Prediction of recidivism in thefts and burglaries using machine learning. Indian Journal of Science and Technology, 13(06), 696-711. https://doi.org/10.17485/ijs t/2020/v13i06/149853

[4] Kiran, P., Mounika, B., Naveen, P., Tejaswini, N., & Karthik, B. (2022). Crime data anlysis using machine learning.

[5] G. Krishna, R. Singh, A. Gehlot, N. Yamsani, S. Kathuria and S. V. Akram, "Enhancing the Cyber-Security of Battery Management Systems for Energy Storage," 2023 IEEE World Conference on Applied Intelligence and Computing (AIC), Sonbhadra, India, 2023, pp. 959-964, doi: 10.1109/AIC57670.2023.10263843.

[6] G. Krishna, R. Singh, A. Gehlot, and S. V. Akram, "An IoT-based predictive model for improved battery management system using advanced LSTM model," J Energy Storage, vol. 101, p. 113694, Nov. 2024, doi: 10.1016/J.EST.2024.113694.

[7] G. Krishna, R. Singh and A. Gehlot, "Cloud-based Monitoring of the Health of Battery using IoT," 2022 5th International Conference on Contemporary Computing and Informatics (IC3I), Uttar Pradesh, India, 2022, pp. 2232-2235,
doi: 10.1109/IC3I56241.2022.10072860.

[8] South Asian Journal of Engineering and Technology, 12(3), 61-68. https://doi.org/10.26524/sajet.2022.12.39

[9] Kshatri, S., Singh, D., Narain, B., Bhatia, S., Quasim, M., & Sinha, G. (2021). An empirical analysis of machine learning algorithms for crime prediction using stacked generalization: an ensemble approach. Ieee Access, 9, 67488-67500. https:// doi.org/10.1109/access.2021.30751

[10] Lamari, Y., Freškura, B., Abdessamad, A., Eichberg, S., & Bonviller, S. (2020). Predicting spatial crime occurrences through an efficient ensemble-learning model. Isprs Interna- tional Journal of Geo-Information, 9(11), 645. https://doi.org/ 10.3390/ijgi9110645

[11] Palanivinayagam, A., Shankar, G., Bhattacharya, S., Anumbe, N., Ibeke, E., & Biamba, C. (2021).

[12] An optimized machine learning Sardana, D., Marwaha, S., Bhatnagar, R., & Sardana,S. (2021). Supervised and unsupervised machine learning methodologies for crime pattern analysis. International Journal of Artificial Intelligence & Applications, 12(1), 83-99.

[13] Diwakar, M., Sharma, K., Dhaundiyal, R., Bawane, S., Joshi, K., & Singh, P. (2021, April). A review on autonomous remote security and mobile surveillance using internet of things. In *Journal of Physics: Conference Series* (Vol. 1854, No. 1, p. 012034). IOP Publishing

[14] Wang, S. (2021). Hate crime analysis based on artificial intelligence methods. E3s Web of Conferences, 251, 01062. h ttps://doi.org/10.1051/e3sconf/202125101062

[15] Wheeler, A. and Steenbeek, W. (2020). Mapping the risk terrain for crime using machine learning. Journal of Quanti- tative Criminology, 37(2), 445-480. https://doi.org/10.1007/s10940-020-09457-7

[16] Yin, J. (2023). Crime prediction methods based on machine learning: a survey. Computers Materials & Continua, 74(2), 4601-4629. https://doi.org/10.32604/cmc.2023.034190

[17] Bhardwaj, G. and Bawa, R. (2022). Assaying the statistics of crime against women in india using provenance and machine learning models. International Journal of Advanced Computer Science and Applications, 13(7). https://doi.org/10.14569/ijac sa.2022.0130760

[18] Espinosa, F. (2020). Prediction of recidivism in thefts and burglaries using machine learning. Indian Journal of Science and Technology, 13(06), 696-711.

[19] N. Kanimozhi, N. V. Keerthana, G. S. Pavithra, G. Ranjitha and S. Yuvarani, "CRIME Type and Occurrence Prediction Using Machine Learning Algorithm," 2021 International Conference on Artificial Intelligence and Smart Systems (ICAIS), Coimbatore, India, 2021, pp. 266-273, doi: 10.1109/ICAIS50930.2021.9395953

[20] Sharma, T., Diwakar, M., Singh, P., Lamba, S., Kumar, P., & Joshi, K. (2021, November). Emotion Analysis for predicting the emotion labels using Machine Learning approaches. In 2021 IEEE 8th Uttar Pradesh Section International Conference on Electrical, Electronics and Computer Engineering (UPCON) (pp. 1-6). IEEE.

[21] Alam, A., Singh, L., Jaffery, Z. A., Verma, Y. K., & Diwakar, M. (2022). Distance-based confidence generation and aggregation of classifier for unstructured road detection. Journal of King Saud University-Computer and Information Sciences, 34(10), 8727-8738.

[22] Arya, C., Tripathi, A., Singh, P., Diwakar, M., Sharma, K., & Pandey, H. (2021, April). Object detection using deep learning: A review. In Journal of Physics: Conference Series (Vol. 1854, No. 1, p. 012012). IOP Publishing.

Authorized licensed use limited to: Pontificia Universidad Catolica Del Peru. Downloaded on September 18,2026 at 14:37:41 UTC from IEEE Xplore. Restrictions apply.